

Sami Hatoum

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Education

- University College London** | *MSc Artificial Intelligence and Imaging* 2025 – 2026
Predicted Distinction
- University of St Andrews** | *BSc Computer Science* 2021 – 2025
First Class Honours, Dean's List

Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL
Frameworks: React, Next.js, Node.js, Express.js, Flask, React Native, REST APIs
Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux
Data & Testing: PostgreSQL, Redis, Jest, Pytest, Locust, Pandas, NumPy

Experience

- PricewaterhouseCoopers (PwC)** | *Software Engineering Intern* Jun 2024 – Aug 2024
- Engineered containerised deployment with **Flask REST API** on **AWS EC2** with auto-scaling across multiple instances. Validated **sub-500ms latency** and **150 requests/second** throughput under load testing with **Locust**
 - Built a **sentiment analysis model** using **PyTorch BERT** to automate customer feedback classification for a retail client. Achieved **91% F1 score** on an imbalanced dataset (3:1 negative to positive), processing **10,000 reviews daily**
 - Implemented **CloudWatch** monitoring dashboards and an automated retraining pipeline that caught a 12% accuracy regression within 48 hours, triggering model rollback before production impact
 - Optimised preprocessing with **pandas** vectorisation, reducing daily batch processing time by **60%**
- PricewaterhouseCoopers (PwC)** | *Software Engineering Intern* Aug 2023 – Sep 2023
- Developed an internal **client onboarding dashboard** in **React** and **Node.js** that replaced a manual spreadsheet workflow for a financial services team, reducing onboarding time from **3 days to 4 hours** across 200+ users
 - Designed and implemented a **PostgreSQL** data model with **role-based access control** across 4 user roles, integrating with PwC's internal **SSO authentication** system via **OAuth 2.0**
 - Wrote **CI/CD pipelines** in **GitHub Actions** with linting, automated test gates, and staged deployments to **AWS**; zero failed releases across the internship after pipeline adoption
 - Conducted **code reviews** on **30+ pull requests** with enforced **branch protection rules** and mandatory approvals, flagging issues in error handling and **SQL** query efficiency before merge
 - Achieved **90%+ code coverage** across **45 unit and integration tests** (**Jest**, **Supertest**) spanning the **REST API** and data access layers, catching 3 regressions before they reached staging

Projects

Database Query Optimiser

- Developed a **PostgreSQL** analysis tool in **Python** that parses **query ASTs**, evaluates **EXPLAIN plans**, and recommends indexes. Reduced query execution time from **2.4s to 420ms** on table scans over **500K rows**
- Reviewed **200+ production queries** across a **five-million-row database** (12 tables), identifying **21 missing indexes** that reduced average query time by **65%** and eliminated most sequential scans
- Deployed on **AWS ECS** with **RDS PostgreSQL**, implementing **CloudWatch** alerts for query latency above 5s and **auto-scaling groups** for batch analysis traffic spikes

Real-Time Collaborative Code Editor

- Built a multi-user code editor in **React** and **Node.js** using **WebSockets** and **Operational Transform** for conflict-free concurrent editing, with syntax highlighting for 12 languages via **CodeMirror**
- Implemented a **Redis** pub/sub layer for horizontal scaling across server instances, supporting **50+ concurrent editors** with sub-100ms synchronisation latency and automatic reconnection on network loss
- Wrote **85+ tests** (**Jest**, **Playwright**) covering OT convergence edge cases and WebSocket fault tolerance; deployed on **AWS ECS** with **Docker** and **GitHub Actions** CI/CD
- Added **JWT authentication**, per-room permission levels (owner/editor/viewer), and a session persistence layer in **PostgreSQL** enabling users to rejoin documents with full edit history intact

Selective Prediction for Medical Image Segmentation (MSc Dissertation)

- Built a selective prediction framework to flag unreliable AI segmentation outputs for human review, reducing silent failures in clinical deployment of **nnU-Net** on pelvic MRI
- Achieved **0.89 Dice score at 85% coverage** on 9-class segmentation (**+0.02 Dice** over full-coverage baseline) using **Monte Carlo Dropout** uncertainty estimation with entropy and mutual information aggregation
- Evaluation pipeline generates accuracy-coverage curves, calibration plots, and per-class reliability diagrams across all cross-validation folds, enabling threshold selection per clinical risk tolerance